

Scope and Manufacture

Namespace as the scope of a count, graded copy number, and some cheap lower bounds
on the assembly of a mind

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Abstract

Assembly theory characterises life by two numbers: an assembly index, counting the joining steps needed to manufacture an object, and a copy number, counting how many of that object are present. The first has no observer and the second has no scope. This note supplies both, using machinery the companion notes already carry, and finds that once they are supplied the two numbers stop being independent.

Scope first. A count of copies presupposes a region in which to count, and if that region is the universe there is no census to take — not a hard count but no fact of the matter. The companion notes work throughout in *namespaces*, and a namespace is exactly a scope. We take one step past the finite-set-of-channels version: a scope should be a *predicate* rather than a set, so that it can be finitely described and yet unbounded in extension. Disjointness of two such predicates is what makes a composite scope parse uniquely, and so is a feature and not an obstacle; and with a fixed-point operator, $\mu X. @((\phi \vee X) \mid (\psi \vee X))$ describes a self-similar scope in a constant number of symbols. The reflective calculus makes membership in such a scope decidable by descent, because a name codes a strictly smaller term.

Copy number second. Bisimilarity is the right identity criterion for copies and is not decidable, so a copy number defined by it is well posed and uneffective. Relativising to the resolution a learner can afford — n -step bisimulation, with n bounded by its own assembly index — makes the count effective and makes it observer-relative. Coarse observers *overcount*, geometrically: in a population separating at depth h with branching b , an observer of resolution n reports b^{h-n} copies where there is one. And in a scope generated by a fixed point the count is not an integer but a series, one term per stratum; assembly theory's single number is the flat collapse of that series, licensed only when the scope has no strata to collapse.

Manufacture third, and lightly. We give six cheap lower bounds on the assembly index of an ecology-as-mind — having an inside, owning a boundary, carrying a germ, resolving at a depth, escaping a basin, and repairing itself — and observe that disjointness of the strata is what lets them be added rather than maximised. None is tight and we do not argue that tightness is yet the interesting question.

Two things follow that we did not expect. Copies are forced by repairability and by sampling before they are wanted for statistics, and the sampling requirement is exponential in depth, so $CN \gtrsim p^{-h}$ for any ecology that actually uses its architecture: the joint biosignature acquires a functional form whose base is medium reliability. And the assembly budget of a multi-skilled mind is dominated by the number of distinct niches it addresses rather than by the depth of its architecture, because self-similar architecture is paid for once. A worked four-skill individual, computed in full, spends 57% of its floor on content and saves 35% of the unshared budget by sharing its kernel.

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1 Introduction

1.1 The question this note asks

The four notes preceding this one move the unit of learning outward. The first [1] makes an individual mortal scientist out of cost-accounted rho and a logic of namespaces. The second [2] plays that individual against an adversary and prices its ladder of hypotheses. The third [3] composes learners by parallel composition and reads the ecological relation between two of them off a grading vector over typed namespaces. The fourth [4] gives each community its own token colour, adjoins engines to convert between them, and finds that error correction is what decides when a composite may close over its namespace and become an individual — at which point the unit has moved far enough to start again.

That series leaves a natural question unanswered: what does all this motion *cost*, and in what region is the cost counted? Assembly theory [13, 14] offers a vocabulary for the first half of that question. It assigns to an object an *assembly index*: the minimum number of joining steps needed to build it from elementary material, where each step may reuse anything already made. And it pairs that index with a *copy number*, on the thesis that high assembly index together with high copy number is a biosignature — that only a process of selection makes many of a hard-to-make thing.

The pairing is attractive and, we think, correct in outline. Our complaint is narrower: the two quantities are asymmetrically specified. The assembly index is a minimum over constructions and so is a property of the object; whatever else is unclear about it, it does not depend on who is looking. Copy number does. To count copies of M one must say *where* to count and *what counts as a copy*, and assembly theory as it stands says neither.

1.2 The move, stated once

A namespace is a scope. That is the whole of the observation, and the rest of this note is what follows from taking it seriously.

If the scope is left implicit, and one tries to make it explicit by saying “the universe”, the count does not become large or uncomputable. It becomes ill-formed: there is no region, no census, no simultaneity in which to take one, and the quantifier ranges over something that is not a collection of locations. If instead the scope is a namespace, three things become available at once. The region is a set of channels, so a count is a count of channels. The identity criterion is bisimulation, so “copy” means behaviourally indistinguishable rather than structurally identical. And — this is the part the companion notes make available and assembly theory cannot — both the region and the criterion are things a *budgeted learner* can hold, which means the count acquires the observer that it needs and that assembly theory forgot to give it.

Once the observer is there, the two numbers couple. The observer’s resolution is bounded by its own manufacture; a coarse observer overcounts; and the copies that a deep architecture requires are forced by the fragility of depth. High assembly index and high copy number are not two independent readings that happen to correlate in living matter. They are two coordinates on a frontier, and we can write down the frontier.

1.3 What we claim, and what we do not

We claim: a definition of scope as a predicate rather than a set (§3); a proposition that disjointness makes composite scopes parse uniquely, so that structure is what disjointness buys (§3.3); fixed-

point scopes and their decidability by descent (§3.4); a graded, resolution-indexed copy number (§4) with a geometric overcounting law; six cheap lower bounds on manufacture (§5); a generator/unfolding distinction with two ratios (§6); and a coupling $CN \gtrsim p^{-h}$ between the two axes of the biosignature (§7).

We do not claim that any lower bound in §5 is tight, or even close. We think they are all loose, in some cases wildly, and we give them anyway: they are cheap, they are honest, and a floor one can compute is worth more at this stage than a floor one can defend. The question of what the *true* minimum manufacture of a mind is seems to us to require an incompressibility argument of a kind nobody currently knows how to make in this setting, and we would rather leave the question visible than paper it over.

We also do not claim that the copy series of §4.3 is the only possible grading, or that the four-skill composite of §7.5 is a model of any actual animal. It is an illustration with the numbers carried through, in the manner of [3, §12], and its purpose is to show that the quantities in this note can be computed at all.

2 What this note carries forward

Enough to orient; the companion notes carry the detail.

From assembly theory. $AI(M)$ is the minimum number of joining steps from elementary material to M , each step joining two objects already present. $CN(M)$ counts copies of M in a sample. The biosignature thesis is that $AI(M) \gg 1$ together with $CN(M) \gg 1$ indicates selection. In the process setting the index has an exact reading: $AI(M)$ is the minimum length of a context-labelled path in the *open* synchronisation tree from the elementary terms to M , because a joining step is precisely a transition that requires the environment to supply a partner, and those are the transitions the open tree records.

From [1]. A computation is ontologically isolated, so its environment is other computations, and the boundary between them is the *cut*. Across the cut there are three grades of access: perception (reading structure the environment has already computed into legible form), interaction (rendezvous), and reflection (quotation and its inverse). Hypotheses are formulae of a namespace logic; an assay is a complementary guard pair whose third outcome is budget-relative refutation. Confinement: an ultrametric walk of steps all shorter than 2^{-n} never leaves the ball of radius 2^{-n} , so a purely incremental learner cannot escape its initial basin.

From [3]. A learner is a distribution over a population together with its resource reservoirs, and that population *is* a namespace. Learners compose by parallel composition. Disjointness of namespaces is graded separation at $k = 0$, and the factorisation theorem says disjointness is equivalent to the absence of a spanning redex and to probabilistic independence of the composed distributions; the defect of the lax comparison map is the interaction. Rooting each learner's namespace at its own quotation makes disjointness a theorem rather than a hope, by freshness-by-quotation.

From [4]. Communication is mediated: a rendezvous is realised by an interposed process, and this is without loss of generality. Media are processes, hence nameable, hence form a namespace $Chn(B)$ one level up the reflection tower; and *the medium namespace of one level is the endpoint*

namespace of the next, which is the generating step of the fractal. An individual is a region whose internal media are closed and whose boundary media are not; perfect closure is death; individuation is a fixed point and the circularity is the fractality. A percept of integration depth d survives with probability p^d and costs d metered rendezvous. The price code on the engine graph has total detection strength β_1 by Foster’s theorem, so a bridge carries no discipline at all.

One theorem, used repeatedly. No process contains a name that codes itself; a name codes a strictly smaller term [7]. This is what makes the reflection tower well-founded downwards, what makes freshness by quotation work, and — new here — what makes membership in a fixed-point scope decidable by descent.

Notation. $@P$ is the name quoting P and $*n$ the process a name carries. $\sim$ is bisimilarity and $\sim_n$ its n -step approximant. $\phi \mid \psi$ is the structural (separating) conjunction at process level and $@\phi$ the name predicate satisfied by $@P$ when $P \models \phi$. $|_k^N$ is graded separation over a described namespace, as in [2, §3].

3 Scope

3.1 The unbound quantifier

Write the copy number as it is usually given:

$$\text{CN}(M) = |\{x : x \text{ is a copy of } M\}|.$$

Two things are unbound. The variable x ranges over — what? And “is a copy of” is — which relation?

These are different failures and they fail differently. The second is a definitional gap: pick a relation and the criterion is fixed, and the argument is then about whether the relation is the right one. The first is worse than a gap. If one answers “the sample”, the number is relative to a sample that is not part of the theory, so two experimenters with different jars report different biosignatures for the same chemistry. If one answers “the universe”, the number is not merely uncomputable.

Remark 3.1 (Universe-as-scope is not a hard count). A count over the universe is ill-formed rather than intractable. It requires a collection over which to range and a moment at which to range over it; general covariance denies the second, and the first is not a set of locations in any sense the theory supplies, since assembly theory has no notion of location at all. The correct diagnosis is not “we cannot evaluate this” but “there is nothing here to evaluate”. This matters because the two diagnoses recommend different repairs. Intractability recommends approximation; ill-formedness recommends supplying the missing argument.

The missing argument is scope, and the companion notes have been working in one all along without naming it as such.

3.2 Scope is a predicate, not a set

The obvious repair is to make the scope a finite set of channels. In the rho calculus a channel is a name $@P$, so a finite set of channels is a perfectly good region: a count over it is a count of channels, and every channel is a location because it is a piece of code at which something resides.

That repair works and it is too weak. Its weakness is that a finite set must be given extensionally — listed — and a scope worth having is usually one nobody can list. “The channels belonging to this organism” is a scope; so is “the channels of any learner rooted at $@P$ ”, and so is “the channels of this organism, and of its organs, and of their cells, and so on down”. None of these is naturally a list, and the last is not finite at all.

Definition 3.2 (Scope). A *scope* is a name predicate N . The *extension* of a scope is

$$\text{Ext}(N) = \{ n : n \models N \},$$

a set of channels, possibly infinite. We write $@\phi$ for the scope satisfied by $@P$ exactly when $P \models \phi$, and read it as “names of ϕ -things”.

Remark 3.3 (The extensional version is the flat case). A finite set of channels $\{c_1, \dots, c_k\}$ is the scope $@\phi_1 \vee \dots \vee @\phi_k$ where ϕ_i is a characteristic formula for $*c_i$. So nothing is lost by the move to predicates; what is gained is that a scope may now be *finitely described and unboundedly large*, which is the case every interesting application is in.

The gain is not cosmetic. A learner’s hypotheses are formulae; a learner’s resources are held at channels; and a learner can only work with a scope it can *state*. Making scope a predicate puts it in the same syntactic category as everything else the learner traffics in, which is what allows the rest of this note to price it.

3.3 Disjointness is structure

A composite scope should be more tractable than its parts, not less. That sounds backwards and is not, and the reason is a small proposition about parsing.

Proposition 3.4 (Unique decomposition). *Let $@\phi$ and $@\psi$ be scopes with $\text{Ext}(@\phi) \cap \text{Ext}(@\psi) = \emptyset$, and suppose ϕ and ψ are separated at grade 0 in the sense of [3, §5]. Then every $n \in \text{Ext}(@(\phi \mid \psi))$ has a unique decomposition: there are P, Q with $n = @(P \mid Q)$, $P \models \phi$, $Q \models \psi$, and P, Q are determined up to structural congruence.*

Proof sketch. Existence is the definition of $|$ at formula level. For uniqueness, suppose $n = @(P \mid Q) = @(P' \mid Q')$ with $P, P' \models \phi$ and $Q, Q' \models \psi$. Quotation is injective, so $P \mid Q \equiv P' \mid Q'$. Grade-0 separation says there is no spanning redex between the parands, so by the factorisation theorem of [3, §5] the free names of the ϕ -part and the ψ -part are disjoint. A structural congruence between two parallel compositions whose parands have disjoint free names permutes parands within each side; hence $P \equiv P'$ and $Q \equiv Q'$. $\square$

Corollary 3.5 (Description is additive under disjointness). *Under the hypotheses of Proposition 3.4, the description length of the composite scope is $\ell(@(\phi \mid \psi)) = \ell(\phi) + \ell(\psi) + O(1)$, and membership testing decomposes: a name is tested by splitting it once and testing each part against its own predicate.*

Remark 3.6 (Overlap is where you pay, and it is the same payment as before). If the scopes overlap, the split is not unique, and a membership test must search over splittings rather than perform one. The description of the composite then does not decompose, and the excess is exactly the defect of the lax comparison map $\lambda: s(\mathcal{L}_1) \otimes s(\mathcal{L}_2) \rightarrow s(\mathcal{L}_1 \mid \mathcal{L}_2)$ of [3, §5]: the interaction between two learners and the ambiguity in parsing their joint namespace are the same quantity, seen once dynamically and once syntactically. This is the third appearance of that pattern in the series, after [3]’s λ and [4]’s θ .

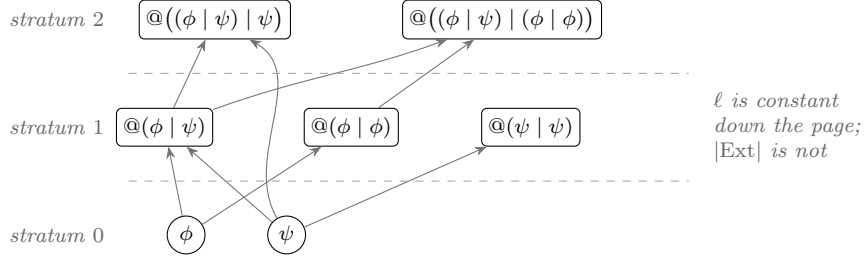


Figure 1: The extension of a generated scope, stratified. Every name at stratum $j + 1$ quotes a parallel composition of drops of names at stratum $\leq j$, and by Proposition 3.4 that decomposition is unique. The generator is three symbols wide at every depth; the extension is not (Table 1).

So disjointness is not a constraint that composition must work around. It is the certificate that makes a composite scope *structured*, and structure is what makes a larger scope cheaper to work in than a smaller unstructured one. We will use this again in §5.4, where the same certificate is what licenses adding lower bounds instead of maximising them.

3.4 Fractality by fixed point

Disjoint composition gives wider scopes. It does not by itself give *deeper* ones — scopes whose members are themselves composites of members, all the way down. For that we want a fixed point.

Definition 3.7 (Generated scope). Let ϕ, ψ be formulae and X a scope variable. The *generated scope*

$$\mathbf{N} = \mu X. @((\phi \vee X) | (\psi \vee X))$$

is the least scope $\mathbf{N}$ with $\text{Ext}(\mathbf{N}) = \text{Ext}(@((\phi \vee \mathbf{N}) | (\psi \vee \mathbf{N})))$, where in the body a scope variable in a process position is read as “the drop of a name in $\text{Ext}(X)$ ”.

More elaborate generators are available and will be wanted — one predicate per skill rather than two, guarded recursion under several variables, gradings carried through the unfolding — but this one is enough to make the points.

Definition 3.8 (Stratum). The *stratum* $\text{str}(n)$ of a name $n \in \text{Ext}(\mathbf{N})$ is the number of unfoldings of μX needed to derive $n \models \mathbf{N}$. Atoms — names of ϕ -things and ψ -things that are not themselves composites — have stratum 0.

The extension grows doubly exponentially while the generator does not grow at all. That gap is the whole point of the construction, and §6 is about what to make of it. First, though, the property that makes generated scopes usable rather than merely compact.

Proposition 3.9 (Membership is decidable by descent). *Let $\mathbf{N}$ be a generated scope whose atomic predicates ϕ, ψ have decidable satisfaction. Then for any name n , the question $n \models \mathbf{N}$ is decidable.*

Proof sketch. Test n by unfolding: $n \models \mathbf{N}$ iff either $*n \models \phi$ or $*n \models \psi$, or $n = @(P | Q)$ with $@P, @Q$ each satisfying $\mathbf{N}$. By Proposition 3.4 the split is unique, so no search is needed. The recursive calls are on $@P$ and $@Q$, and by the no-self-code theorem a name codes a strictly smaller term, so $|P|, |Q| < |P | Q|$. The recursion is therefore well-founded on term size and terminates, with depth bounded by $\text{str}(n) \leq |*n|$. $\square$

stratum $\leq j$	$ \text{Ext}_{\leq j}(\mathbf{N}) $	new at this stratum
0	2	2
1	5	3
2	17	12
3	155	138
4	12,092	11,937
5	73,114,280	73,102,188

Table 1: Extension of the two-atom generator of Definition 3.7, counted up to structural congruence: $u_{j+1} = 2 + \binom{u_j+1}{2}$. Computed in Appendix B, block (1).

Remark 3.10 (What is doing the work). Two hypotheses carry this, and neither is decoration. Disjointness makes the split unique, so the descent is deterministic rather than a search; that is Proposition 3.4, and without it the decision procedure branches over splittings and the complexity is no longer governed by term size. No-self-code makes the descent well-founded; that is a property of the reflective calculus specifically, and it is what prevents a name from being a member of a scope by way of itself. So the computational coherence of scope here is not merely a matter of restricting to finite regions. It comes from reflection.

Remark 3.11 (μ , not ν , and this is a claim). Definition 3.7 takes the least fixed point, and the choice is not innocent. The greatest fixed point admits non-well-founded scopes — names whose membership is witnessed only by an infinite descent — and Proposition 3.9 does not cover them, because the argument from no-self-code bounds the descent by term size and a non-well-founded witness has no such bound. We take the view that the framework only ever wants μ , and that this is a small result rather than a convention: *a scope a learner can survey is one whose membership terminates*, and by [1, §6] a learner that cannot terminate a membership test cannot afford the assay that depends on it. A ν -scope is a scope nothing budgeted can inhabit knowingly. We record the alternative as an open problem (§9) rather than as a settled matter, since a non-well-founded scope is precisely what one would reach for to model an ecology with no bottom, and we would rather not foreclose that by fiat.

3.5 Scope is an economic quantity

Definition 3.2 makes scope statable and Proposition 3.9 makes membership decidable. Neither makes a survey affordable, and it is the survey that a copy number requires.

Definition 3.12 (Survey and affordable scope). A *survey* of a scope $\mathbf{N}$ at resolution n is the process of visiting each $c \in \text{Ext}(\mathbf{N})$ and deciding $*c \sim_n P$. Its cost is one metered rendezvous per visit plus the cost of the depth- n decision, which by [2, §4] is $\kappa(n)$ per candidate on the pricing schedule there. The *affordable scope* of a learner with budget σ is the largest $\text{Ext}_{\leq j}(\mathbf{N})$ whose survey costs at most σ .

Proposition 3.13 (Affordable scope is bounded by the profitable radius). *A learner’s affordable scope is contained in the ball of the profitable radius of [4, §7]. Beyond that radius a channel’s traffic cannot repay the metering and loss of reaching it, so visiting it is a strictly negative transaction and a learner that performs it is worse off than one that does not.*

Remark 3.14 (This is what “computationally coherent” should mean). Three conditions, and all three are needed. A scope must be *statable* (a predicate a learner can hold), *decidable* (a

membership test that terminates), and *surveyable* (an extension a budget can visit). Universe-as-scope fails all three; a finite list of channels satisfies all three but expresses almost nothing; a generated scope satisfies the first two by construction and the third stratum by stratum. So there is a canonical thing to do when a scope is too big: truncate the unfolding. That is not an approximation to a true count over a true region. It is the count, taken in the region that exists for this learner.

4 Copy number, graded

4.1 Well posed is not effective

With a scope in hand the identity criterion is easy to state. Two processes are copies when no experiment distinguishes them, and the relation that says so is bisimilarity. This is the right criterion and not merely a convenient one: assembly theory’s implicit appeal to structural identity is at once too strict, since two conformers of a molecule are the same reagent, and too permissive, since two graphs may agree while the things they denote behave differently under interaction. Behavioural identity is what a count of copies was always trying to track.

$$\text{CN}(P, \mathbf{N}) = |\{c \in \text{Ext}(\mathbf{N}) : *c \sim P\}|.$$

This is well posed. It is also not effective: bisimilarity is undecidable for calculi of this expressiveness [12], so no learner computes it, and in particular no learner computes this count. The gap is the same one [1] meets in the assay, where a hypothesis that is true but unaffordable is refuted *relative to a budget*, and the same remedy applies.

Definition 4.1 (Copy number at a resolution). For a process P , a scope $\mathbf{N}$ and $n \in \mathbb{N}$,

$$\text{CN}_n(P, \mathbf{N}) = |\{c \in \text{Ext}(\mathbf{N}) : *c \sim_n P\}|,$$

where $\sim_n$ is n -step bisimulation. By the Hennessy–Milner characterisation [9], $*c \sim_n P$ exactly when $*c$ and P agree on all formulae of modal nesting depth at most n .

Proposition 4.2 (Effectivity). $\text{CN}_n(P, \mathbf{N})$ is computable for finitely branching processes, any generated scope $\mathbf{N}$, and any finite n , at cost linear in $|\text{Ext}_{\leq j}(\mathbf{N})|$ and in the depth- n decision procedure.

The resolution n is not a free parameter. It is the depth of modal nesting a learner can afford to evaluate, and that quantity is bounded — by integration depth, hence by tower height, hence, if one accepts the argument of [6, Ch. 44], by the learner’s own assembly index. So a count of copies has three arguments and the third belongs to whoever is counting.

4.2 The observer enters the count

Proposition 4.3 (Coarse resolution overcounts). For $n' \leq n$, $\text{CN}_{n'}(P, \mathbf{N}) \geq \text{CN}_n(P, \mathbf{N})$, and $\text{CN}_n \downarrow \text{CN}$ as $n \rightarrow \infty$ for image-finite processes.

Proof. $\sim_n \subseteq \sim_{n'}$ for $n' \leq n$, so the counted set shrinks with n ; the limit is the standard approximation theorem for bisimilarity on image-finite processes. $\square$

The direction is worth dwelling on, because it is the opposite of the one intuition supplies. A weak instrument does not miss copies. It *manufactures* them, by failing to separate things that differ below its resolution. And the manufacture is not a small correction.

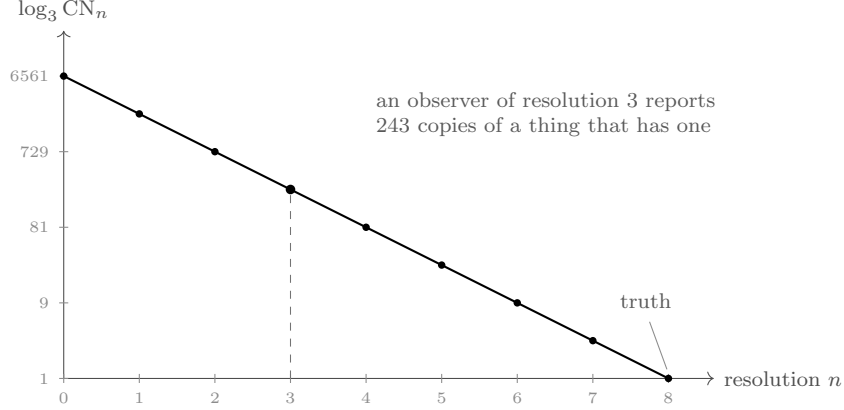


Figure 2: Proposition 4.4 for $b = 3$, $h = 8$; the vertical axis is logarithmic, so the overcount is the vertical gap to the right-hand endpoint. Values in Appendix B, block (2).

Proposition 4.4 (Geometric overcounting). *Let a population be the leaves of a behaviour tree of depth h and branching factor b , with two members n -step bisimilar exactly when their paths agree for n steps. Then for any member P ,*

$$\text{CN}_n(P, \mathbf{N}) = b^{h-n}, \quad 0 \leq n \leq h,$$

so an observer whose resolution falls short of h by k reports b^k times too many copies.

Proof. The members n -step bisimilar to P are exactly the leaves of the depth- n subtree containing P , of which there are b^{h-n} ; at $n = h$ only P itself remains. $\square$

Corollary 4.5 (A detector has a floor of its own). *To report CN correctly for a population whose members separate at depth h , an observer must afford resolution h , and therefore must itself have been assembled to a depth supporting modal nesting h .*

Remark 4.6 (How far this claim goes, and how far it does not). It is tempting to summarise Corollary 4.5 as “it takes a mind to find a mind”, and we think that summary is wrong, or at least much stronger than what is shown. What the observer must match is the depth at which the *kinds* in the population separate, not the assembly index of any member of it. Those are very different numbers: in the worked composite of §7.5 the four kinds separate by depth 4 while the individual carrying them has an assembly floor of 28. A shallow instrument can therefore certify kind structure it could not begin to build. What it cannot do is report a copy number that means anything, and what it will do instead — silently, with no error signal — is report one that is too large by a factor exponential in its shortfall. That is the useful claim, and it is a claim about *instruments* rather than about minds.

Remark 4.7 (A false-positive mechanism for the biosignature). Assembly theory reads $\text{AI} \gg 1$ together with $\text{CN} \gg 1$ as evidence of selection. Proposition 4.4 says the second conjunct is inflated by any instrument that cannot resolve the population, and inflated most where the population is most diverse — which is to say, exactly where selection has been most active in generating variants that a coarse instrument pools. The failure mode is therefore not random noise but a bias aligned with the signal being sought. We do not know how large the effect is in the mass-spectrometric practice the theory was designed around, and we would want that estimated before the criterion is applied to anything as consequential as an extraterrestrial sample.

4.3 The count has a grading it should have kept

In a generated scope there is a second thing wrong with reporting an integer.

Definition 4.8 (Copy series). For a generated scope $\mathbf{N}$ with strata str , the *copy series* of P at resolution n is

$$\text{CN}_n^\bullet(P, \mathbf{N}) = (\text{CN}_n^{(0)}, \text{CN}_n^{(1)}, \text{CN}_n^{(2)}, \dots), \quad \text{CN}_n^{(j)} = |\{c \in \text{Ext}(\mathbf{N}) : \text{str}(c) = j, *c \sim_n P\}|.$$

Equivalently, the generating function $\sum_j \text{CN}_n^{(j)} x^j$.

Remark 4.9 (The integer is the flat collapse). $\text{CN}_n(P, \mathbf{N}) = \sum_j \text{CN}_n^{(j)}$, so nothing is lost in passing to the series and a great deal is lost in passing back. The collapse is legitimate exactly when the scope has one stratum — when it is a flat region of like things, which is the situation a jar of reagent approximates and an organism does not. Assembly theory’s copy number is the flat collapse, and the reason it has felt unproblematic is that the theory’s home application really is a jar.

Remark 4.10 (This agrees with something already in print). [4, §8] concludes, from an entirely different argument about typed namespaces and the medium tower, that the grading vector k is not a vector but a *sequence* of vectors, one per stratum. That is the same observation reached from the composition side rather than the counting side, and the agreement is some evidence that the stratification is a feature of the subject rather than of either argument.

Remark 4.11 (What the series is for). Three uses, in increasing order of speculativeness. First, it distinguishes architectures that the integer confuses: a colony of 1000 like cells and an organism of 1000 cells in four tissues have the same CN and different series. Second, the ratio between consecutive terms is a branching factor, and §7 argues that this factor is forced from below by repair and sampling, so the series has a shape one can predict and check. Third, if the biosignature is to be read off a series rather than a number, the criterion becomes a statement about the shape — and “many copies at low strata, few at high, with a ratio bounded below by reliability” is a much more specific fingerprint than “CN is large”. We do not develop the third here.

4.4 A worked series

Take the two-atom generated scope of Definition 3.7 and populate it: let ϕ -atoms be sources and ψ -atoms be foragers, and let the population instantiate every shape at strata 0 through 2 with multiplicity 12. Then $|\text{Ext}_{\leq 2}| = 17$ shapes (Table 1), the shapes are distributed 2 : 3 : 12 over the strata, and the populated scope has 204 channels.

Ask for CN of a stratum-1 shape, say $@(\phi \mid \psi)$. The truth is 12. A learner of resolution 2 separates all three stratum-1 shapes and reports 12. A learner of resolution 1 separates $@(\phi \mid \psi)$ from $@(\phi \mid \phi)$ but not from $@(\psi \mid \psi)$ if the two atoms agree on their first step, and reports 24. A learner of resolution 0 reports 36, and a learner of resolution 0 that also fails to separate strata — which is what happens when the instrument counts occurrences rather than channels — reports 204.

The series makes the difference legible where the integer does not:

$$\text{CN}_2^\bullet = (0, 12, 0), \quad \text{CN}_1^\bullet = (0, 24, 0), \quad \text{CN}_0^\bullet = (0, 36, 0), \quad \text{flat} = 204.$$

Every one of these is a defensible answer to “how many copies are there”. Only the first is an answer to “how many copies are there, at the resolution at which copies are a kind”.

5 Manufacture: some cheap lower bounds

5.1 What a lower bound would have to be

Upper bounds on what an assembled thing can do are easy in this framework and we have several. Exhibit an invariant that cannot exceed the assembly index — tower height will do — and then bound capability by the invariant. Lower bounds run the other way and are much harder, because AI is a *minimum* over constructions, so a lower bound is the claim that no shorter construction exists. That is an incompressibility statement, and incompressibility statements are notoriously resistant.

There is one tractable route. Find quantities $\iota_1, \dots, \iota_m$ that are (i) monotone along assembly edges, (ii) forced to a minimum value by the capability in question, and (iii) witnessed by *disjoint* sets of edges. Then

$$\text{AI}(E) \geq \sum_i \iota_i(E),$$

and without (iii) one gets only $\text{AI}(E) \geq \max_i \iota_i(E)$. Condition (iii) is the hard one in general and, as §5.4 observes, is exactly what the disjointness of §3.3 certifies in the cases we care about.

We now give six such quantities. They are all obvious once stated, none requires an argument longer than a paragraph, and we make no claim that their sum is anywhere near the truth.

5.2 Six floors

	capability	floor	from
L1	having an inside	$\text{AI} \geq 1$	[4, Def. 5.11]
L2	owning a boundary	$\text{AI} \geq 2$	[4, Prop. 5.12]
L3	carrying a germ	+1	no-self-code
L4	resolving at depth n	$\text{AI} \geq n$	[6, Thm. 44.4]
L5	escaping a basin of radius 2^{-n}	$\text{AI} \gtrsim n$	[1, Prop. 12]
L6	repairing itself	$\text{AI} \geq \beta_1$	[4, Thm. 6.3]

Table 2: Cheap floors on the assembly index of an ecology-as-mind. Each is a paragraph in §5.2; none is tight.

L1: having an inside. An ecology-as-mind is a region distinguished from its environment, so there is at least one medium whose endpoints straddle the boundary. A medium is a process and must be constructed — by no-self-code it is not among the terms it carries — so its construction is a joining step. $\text{AI} \geq 1$.

L2: owning a boundary. Definition of individual in [4, §5.5]: internal media closed, boundary media open. That requires the two to be *distinguishable*, hence separately constructed, since a single medium serving both roles would put the region’s interior under external naming and dissolve the closure that makes repair possible. $\text{AI} \geq 2$. Note that L1 and L2 are not independent of the requirement that the boundary be porous: by [4, Prop. 5.14] perfect closure is death, so a mind that has paid for L2 must also have paid for at least one redex pathway across the boundary, which we fold into L2 rather than counting separately.

L3: carrying a germ. The germ of [1, §11] is a quotation $@P$, transmissible and inert until dropped. By no-self-code, $@P$ is not among the names occurring in P : the germ sits strictly above what it encodes. So the machinery that manipulates germ material occupies a level of the tower above the machinery it manipulates, and by the tower-height bound each level costs at least one edge of a minimum path. Reproduction is $+1$, unconditionally, over the floor of a thing that does not reproduce.

L4: resolving at depth n . This one is free: it is the contrapositive of a bound already proved. If an ecology of assembly index AI cannot distinguish environments that are AI-step bisimilar, then an ecology that *does* distinguish environments differing only at depth n has $\text{AI} \geq n$. Every upper bound on capability is a lower bound on manufacture, read backwards, and this is the cleanest instance.

L5: escaping a basin. By confinement [1, Prop. 12], a walk whose steps are all shorter than 2^{-n} never leaves the ball of that radius. Escape requires a revision move of distance at least 2^{-n} , which by [1, Def. 14] means a locus at term-depth at most n and an operation applied there. Addressing a locus at depth n requires a one-hole context of that depth, which must be built. So a learner that can leave its initial basin pays on the order of n steps for the addressing alone.

L6: repairing itself. By Foster’s theorem on the engine graph [4, §6], the total detection strength of the price code is exactly β_1 , and an edge that lies on no cycle has effective resistance 1 and carries no discipline. So a region that can detect a single-edge fault has $\beta_1 \geq 1$, and one that can detect m independent faults has $\beta_1 \geq m$. Each independent cycle requires an edge that closes it, and those edges are joining steps. $\text{AI} \geq \beta_1$.

5.3 Genetic exploration, and the difference between a learner and a scorer

L3 and L5 combine into the one bound in this section we find genuinely interesting, because it separates two things the framework has otherwise been treating together.

Proposition 5.1 (A learner costs strictly more than a scorer). *Let S be an ecology that evaluates hypotheses drawn from a fixed ball of radius 2^{-n} and never leaves it, and let L be one that can leave. Then*

$$\text{AI}(L) \geq \text{AI}(S) + 1 + n^-,$$

where the $+1$ is the germ level of L3 and n^- is the term-depth of the shallowest locus whose revision leaves the ball.

Proof sketch. By confinement, no sequence of moves available to S leaves the ball, so whatever L has that S lacks is not a longer run of the same machinery. By [1, §6.5] the escape available inside this framework is recombination, which requires germ material; by L3 germ material costs a tower level; by L5 the move must address a locus at depth n^- . The germ level and the addressing contexts are constructed from different material — the first from the learner’s own code as quoted, the second from the term structure being addressed — so their edges are distinct. $\square$

Remark 5.2 (What this says about genetic exploration). Read as a design statement rather than a bound: sexual recombination is not an optimisation of an existing search, it is the purchase of a search that was not previously available at any price. The confinement result says an individual cannot get out of its basin by trying harder; Proposition 5.1 says what getting out costs, and the

cost is structural — a level of the tower — rather than metabolic. That is a satisfying place for the price of sex to sit, because it explains why the price is paid once, architecturally, rather than continuously.

Remark 5.3 (The suite of skills does not multiply this). A mind with k skills, each realised as an ecology-as-mind, does not pay L3 k times. The germ is a quotation of the whole individual’s code, and quotation does not distribute over the parts in a way that would require one germ level per skill. This is the first appearance of the economy that §6 and §7.5 make quantitative: architecture is paid for once, content is paid for per skill.

5.4 Disjointness is the additivity certificate

Proposition 5.4 (Disjoint strata have disjoint witnesses). *Let an ecology E have strata rooted at pairwise disjoint scopes $N_1, \dots, N_m$ in the sense of §3.3. Then no single joining step contributes to two strata, and therefore $\text{AI}(E) \geq \sum_i \iota(N_i)$ for any per-stratum floor ι satisfying (i) and (ii) of §5.1.*

Proof sketch. A joining step produces a term T ; the step contributes to stratum i only if $@T \in \text{Ext}(N_i)$. Disjointness of the scopes makes that membership exclusive, so each step is charged to at most one stratum, and the per-stratum floors are supported on disjoint edge sets of any construction path — in particular of a minimum-length one. $\square$

Remark 5.5 (The obligation this discharges). The tower-height bound of [6, Ch. 44] leaves open whether the joining steps for distinct levels are distinct edges of a single minimum-length path, noting that if one construction could serve two levels the count would collapse. Proposition 5.4 discharges that obligation whenever the levels are rooted at disjoint scopes — which, by the rooting discipline of [3, §9], is something a designer or a lineage can arrange rather than something that must be hoped for. The general case, where levels are not disjointly rooted, remains open, and we suspect it is genuinely harder rather than merely unwritten.

5.5 Why we are stopping here

The sum of Table 2 for a reproducing, repairing, depth- n -resolving individual is something like $n + \beta_1 + 4$, and the worked composite of §7.5 evaluates it at 28. We believe the true minimum is very much larger, for the reason [6, Rem. 44.2] gives: the assembly index counts everything built, including redundancy, repair machinery and bulk, while these floors count only the building that buys a named capability. The ratio between them is the interesting quantity and we cannot compute it.

We think that is a reasonable place to leave it. A framework earns attention by making quantities *visible* before it makes them *sharp*, and a floor that anyone can check and improve is a better invitation than a theorem that closes the question. The lower bound problem for minds is, on our reading, at about the stage that the lower bound problem for circuits was before anyone had a technique: everyone can see what a bound would have to look like and nobody can produce one. We would be pleased to be scooped on this.

6 Generator and unfolding

Table 1 shows a generator of fixed size whose extension grows doubly exponentially. That gap deserves a name, because the two quantities on either side of it are measuring different things and

the framework has been conflating them.

Definition 6.1 (Generator length). The *generator length* $\ell(\mathbf{N})$ of a generated scope is the number of symbols in the μ -formula defining it, with atomic predicates counted by their own lengths.

Proposition 6.2 (The chain). *For an ecology E whose namespace is generated by $\mathbf{N}$,*

$$\ell(\mathbf{N}) \leq h(E) \leq \text{AI}(E)$$

under the convention that atomic predicates are at least one symbol and each level of the tower is generated by at least one unfolding.

Remark 6.3 (Three quantities, three questions). ℓ answers “how long is the description”; h answers “how many levels of inside does it have”; AI answers “how many steps did it take to build”. Under μ , ℓ is constant while h grows with the unfolding and AI grows at least as fast as h . Two ratios follow, and they measure different virtues:

$$\frac{h}{\ell} = \text{how much depth a description buys}, \quad \frac{h}{\text{AI}} = \text{architectural efficiency}.$$

The second is the ratio [6, Rem. 44.2] proposes as a measure of how much of an organism’s manufacture bought a new inside. The first is new here, and it is the one that corresponds to something biologists already measure: a small genome producing a deep organism is a large h/ℓ .

Remark 6.4 (The germ is the generator). This is not an analogy, and the framework has both objects already. The germ of [1, §11] is $@P$: inert, transmissible, inspectable, and costing nothing until dropped with $*$. The generator of Definition 3.7 is a formula: finite, transmissible, and costing nothing until unfolded. In an ecology whose namespace is generated, these coincide — what is inherited is the generator, and development is the unfolding. The assembly index prices the unfolding, which is why a genome can be small while its organism is deep, and why the two numbers were never going to be the same number.

Remark 6.5 (A caution about the direction of the depth bound). The bounds of [6, Ch. 44] are additive: one level, one edge, one unit of modal depth. That additivity is inherited from the tower argument and we have no reason to doubt it there. But a lower bound routed through characteristic formulae rather than through the tower would need to know how much the modal depth of $\chi_{P|Q}$ can exceed those of χ_P and χ_Q , and if a single joining step can more than increment it then bounds of that shape degrade from n to $\log n$. We have not needed such a route here — all six floors of §5.2 go through structure rather than through χ — but anyone extending this work should check it before relying on it. We record it in §9.

7 The pressure on copy number

The two parameters of the biosignature look independent. One is a property of an object and the other a property of a population, and nothing in assembly theory relates them except the empirical observation that in living matter both are large. In this framework they are related, and by two mechanisms that both push in the same direction: an architecture that is deep *requires* copies before anybody wants them for statistics.

7.1 Detection forces copies

By Foster’s theorem on the engine graph [4, §6], the total detection strength of the price code is β_1 , distributed over edges by effective resistance, and an edge on no cycle is undetectable. A region whose internal media form a tree therefore has no error correction at all: every fault is silent, and by [4, Cor. 7.3] a bridge fails twice, having neither a cycle to check it nor an owner to repair it.

So repairability requires cycles, and a cycle requires a redundant pathway, and a redundant pathway between two functional positions is two copies of a relation where one would carry the traffic. The requirement recurs per stratum, because each level of the tower has its own media and its own faults.

Proposition 7.1 (Repair floor). *An ecology whose stratum- j media can detect m_j independent single-edge faults has $\beta_1^{(j)} \geq m_j$, and therefore carries at least m_j media in excess of a spanning structure at that stratum.*

This is a floor on copies of *media*, not of learners, and it is small — one per independent fault. The second mechanism is not small.

7.2 Sampling forces copies, exponentially

By [4, Prop. 5.18], a percept of integration depth d traverses d hops of internal medium and survives with probability p^d . A hypothesis at that depth needs m independent confirmations before a learner will fund a revision on it, and confirmations that are attempted in parallel are attempted by *copies*.

Proposition 7.2 (Sampling floor). *To obtain m usable depth- d confirmations per round, an ecology requires*

$$\text{CN} \geq \lceil m p^{-d} \rceil$$

learners positioned at that depth.

$p \backslash d$	1	2	3	4	5	6	7	8
0.99	21	21	21	21	22	22	22	22
0.95	22	23	24	25	26	28	29	31
0.90	23	25	28	31	34	38	42	47
0.80	25	32	40	49	62	77	96	120
0.70	29	41	59	84	119	170	243	347
0.50	40	80	160	320	640	1280	2560	5120

Table 3: $\lceil m p^{-d} \rceil$ for $m = 20$: copies needed to sustain 20 confirmations at integration depth d on media of reliability p . Appendix B, block (3).

7.3 The two axes are coupled

Proposition 7.3 (The frontier). *An ecology that uses its architecture — that is, one whose hypotheses are evaluated at integration depths reaching its tower height h — has*

$$\text{CN} \gtrsim m p^{-h},$$

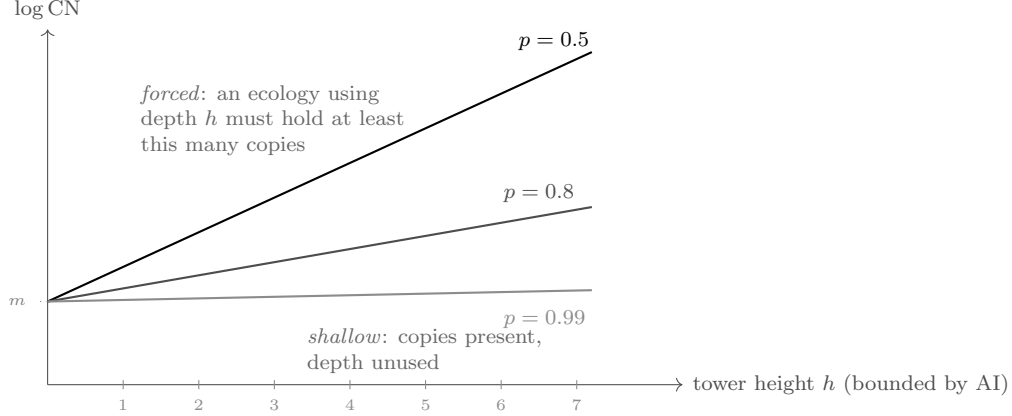


Figure 3: Proposition 7.3. Each line is $\log \text{CN} = \log m + h \log(1/p)$: the copy number an ecology must hold to use a tower of height h on media of reliability p . The slope is set by reliability alone. Assembly theory’s joint biosignature is the claim that living matter sits far to the upper right; this framework says the two coordinates are not free of each other once the architecture is in use.

and since $h \leq \text{AI}$, the reachable region of the (AI, CN) plane at reliability p lies above the curve $\text{CN} = mp^{-\text{AI}}$ only for ecologies that do not use their full depth.

Remark 7.4 (What the biosignature becomes). “High AI and high CN” is, on this reading, not a conjunction of two independent findings but a point above a line whose slope is $\log(1/p)$. The framework accordingly predicts something the bare conjunction does not: the *ratio* $\log \text{CN}/h$ should cluster near $\log(1/p)$ for systems that are using their architecture, and fall well below it for systems that are merely abundant. A crystal is abundant and shallow; a bacterium is deep and abundant in the proportion its chemistry can sustain. If the ratio is measurable, this is falsifiable, and it is falsifiable in a way the conjunction is not.

7.4 A prediction about scarcity

Depth is priced exponentially and width linearly. Under a contraction of resources, a lineage should therefore shed the exponentially priced axis first.

Remark 7.5 (Depth goes before population). An ecology entering scarcity may reduce h or reduce CN. Reducing h by one releases a factor p^{-1} of the sampling requirement at every stratum above the cut, together with the media, repair and engine machinery of the level removed; reducing CN by one releases one learner’s maintenance. The first is worth vastly more per unit of structural change. So we expect — as an empirical matter, and stated as an expectation rather than a theorem — that lineages entering sustained scarcity lose architectural levels before they lose population, and that the reverse pattern indicates something other than resource pressure. Regressive evolution in parasites and cave fauna is the obvious place to look, and the obvious confound is that the same pressure removes the selective reason for the level as well as the means to pay for it.

7.5 A worked composite: four skills, one individual, twelve of them

We now carry the numbers through a mind whose skills are themselves ecologies-as-mind, in the manner of [3, §12]. The illustration is orca-shaped, and the shape is doing no work beyond making

the skills memorable: a hunting skill, a skill for one’s place in the pod, a mating skill, and a skill for rearing calves. Each is a population of mortal scientists in its own typed namespace with its own token colour; the four are wired by engines; the whole closes over its namespace and is an individual; and twelve such individuals form a pod, which is a candidate individual one stratum up.

The generator. The individual’s scope is

$$N = \mu X. @((\phi_{\text{for}} \vee X) \mid (\phi_{\text{pod}} \vee X) \mid (\phi_{\text{mat}} \vee X) \mid (\phi_{\text{rea}} \vee X)),$$

the four-predicate version of Definition 3.7. The four predicates are pairwise disjoint — each skill’s population is rooted at its own quotation, per [3, §9] — so Proposition 3.4 applies and the scope parses uniquely, Proposition 3.9 makes membership decidable, and Proposition 5.4 lets the floors be added.

Depths and populations. Each skill sits on media of its own reliability: the pod and rearing skills run on internal, owned media ($p = 0.99$); hunting reaches across the boundary into a noisy world ($p = 0.95$); mating runs across *another individual’s* boundary and is worst ($p = 0.90$). Optimal integration depth follows [4, §5.5], maximising $Y(d)p^d - cd$ with $Y(d) = 100(1 - 2^{-d})$ and $c = 2$; the sampling floor follows Proposition 7.2 with $m = 20$.

skill	p	d^*	net value	CN floor	separates at
Forage	0.95	3	69.0	24	4
Pod	0.99	5	82.1	22	2
Mate	0.90	3	57.8	28	3
Rear	0.99	5	82.1	22	3
96					

Table 4: The four skill-ecologies. “Separates at” is the modal depth at which that skill’s learners become distinguishable from the shared kernel behaviour. Appendix B, block (4).

Note which skill is most expensive in copies. It is not the deepest; it is the one on the worst medium. Mating costs 28 learners to sustain depth 3 while the pod skill costs 22 to sustain depth 5, because reliability enters the floor exponentially and depth only through the exponent. An architecture is expensive where it is *unreliable*, not where it is deep.

The copy series. Twelve individuals, each with 96 learners in four skill-ecologies:

$$\text{CN}^\bullet(\text{pod}) = (1152, 48, 12, 1),$$

being learners, skill-ecologies, individuals, and the pod itself. The flat collapse is 1213, a number that answers no question anyone would ask. It is worth saying plainly what the series says and the integer does not: the ratios 1152:48:12:1 are 24, 4, 12, and those three numbers are respectively a sampling floor, a skill count, and a group size — three different kinds of fact about the animal, which the collapse blends into one meaningless total.

component	steps	from
kernel: inside, boundary, germ, stack, assay	5	L1–L3, paid once
differentia: d^* per skill, $3 + 5 + 3 + 5$	16	L4, L5
wiring: joining four parts, $\lceil \log_2 4 \rceil$	2	
engines: four, to close a cycle over four colours	4	L6
closure of the individual’s namespace	1	L2
floor	28	
same, with no kernel sharing (5×4)	43	

Table 5: The assembly floor of the four-skill individual.

The assembly floor. Applying Table 2 with the additivity of Proposition 5.4:

Two numbers to take from this. Sharing the kernel across the four skills saves 15 of 43 steps, or 34.9%; and of the 28 that remain, 16 — 57.1% — are *content* rather than architecture. The fractal is cheap and the niches are expensive, which is the quantitative form of the claim in §5.3.

Remark 7.6 (The economy runs the other way from intuition). One expects a mind made of minds to be expensive because it is deep. It is not: depth is what the fixed point gives away for free, since a generator that describes one level describes all of them and copies of a made thing are free in assembly index. What a multi-skilled mind pays for is *differentiation* — the distinct predicates, assays and token colours that make hunting not-mating. Under this accounting an animal’s assembly index tracks the number of distinct niches it addresses far more closely than the elaborateness of its architecture, and two animals of very different apparent sophistication addressing the same number of niches should come out close together.

Does the whole close? By [4, §7] a region may close over its namespace iff its isoperimetric ratio clears $(\varepsilon\theta/\rho)r$. Taking $\theta = 0.2$, $\rho = 5$, and $\varepsilon = 1 - p$ for the relevant media: the individual, with internal media at $p = 0.99$ and two of its four skills facing outward ($h = 0.5$), has $\varepsilon\theta/\rho = 4 \times 10^{-4}$ and a critical radius $r^* = 1250$ — it closes with enormous margin. The pod, whose media run across individual boundaries at $p = 0.90$ and whose isoperimetric ratio we take as 0.3, has $\varepsilon\theta/\rho = 4 \times 10^{-3}$ and $r^* = 75$. A pod of twelve closes. An aggregation of two hundred does not.

Remark 7.7 (A critical group size, and how much to believe it). The number 75 is a computed consequence of parameters we invented, and nothing in this note determines θ , ρ or the isoperimetric ratio of a pod. What is not invented is the *shape* of the result: there is a finite size beyond which a social aggregation cannot close over its namespace, it scales as $\rho/(\varepsilon\theta)$, and it is set by the reliability of the media running between individuals rather than by anything about the individuals. We note without pressing it that the literature on cohesive group size in social mammals lives in this neighbourhood, and that the framework predicts the bound should move with communication reliability rather than with brain size. We would not want that repeated as a result.

What a passing instrument sees. Finally, the point of §4.2, made concrete. The four skills’ learners share the kernel and separate at depths 2, 3, 3, 4. An observer surveying the pod’s namespace at resolution n reports:

resolution n	kinds	largest class	
0	1	1152	everything is one kind
1	1	1152	
2	2	888	the pod skill separates
3	4	336	mating and rearing separate; foraging is the remainder
≥ 4	4	336	truth

Table 6: An observer’s report on the learner stratum, by resolution. Truth is four kinds, largest class 336.

An instrument at resolution 0 reports a copy number 3.4 times the truth and a monoculture where there are four kinds. At resolution 2 it is still wrong by a factor of 2.6 on the largest class. And note the small mercy at $n = 3$: once every kind but one has been separated, the last comes free, because what remains in the pool is a kind by elimination. A survey therefore needs resolution equal to the second-deepest separation, not the deepest — a saving of exactly one kind’s worth of instrument, and the only place in this note where the accounting is generous.

8 What is shown and what is not

Shown, in the sense that the argument is complete given the companion notes. That a scope may be a predicate rather than a set, and that the extensional version is the flat special case (§3.2). That disjointness makes a composite scope parse uniquely and its description additive (Proposition 3.4, Corollary 3.5). That membership in a μ -generated scope is decidable by descent, because no-self-code bounds the descent by term size (Proposition 3.9). That copy number relativised to resolution is effective, monotone, and geometrically inflated by coarse instruments (§4.2). That in a stratified scope the count is a series and the integer is its collapse (§4.3). That the six floors of §5.2 hold, and may be summed when the strata are disjointly rooted (Proposition 5.4).

Argued, but resting on a parameter we cannot fix. The frontier $CN \gtrsim mp^{-h}$ (Proposition 7.3) depends on p , the per-hop reliability of internal media, which is a free parameter of [4] and is not measured anywhere. Everything downstream of it — the slope in Figure 3, the sampling column of Table 4, the critical group size — inherits that freedom.

Illustrated only. The four-skill composite. Its structure follows the framework; its numbers follow from parameters chosen to be plausible rather than measured. It shows that the quantities are computable and roughly what they look like. It shows nothing about orcas.

Not shown, and we think importantly so. That any floor here is tight, or within an order of magnitude of tight. That the least fixed point is the right one in general (Remark 3.11). That the CN inflation of Remark 4.7 is large in the mass-spectrometric practice assembly theory actually uses — we have given a mechanism, not an estimate. And that any of this is the right way to read Cronin and Walker’s proposal rather than a different proposal wearing its vocabulary; on that we are content to have made the two readings comparable.

9 Open problems

1. **ν -scopes.** Remark 3.11 argues that a budgeted learner can only inhabit a μ -scope, since membership must terminate. But an ecology with no bottom — turtles all the way down — is exactly what one would model with ν , and the reflection tower is bounded below only by no-self-code, which is a fact about names and not about ecologies. Is there a coherent notion of a scope that is non-well-founded and yet surveyable stratum by stratum?
2. **Does one joining step increment modal depth?** Remark 6.5. If $\chi_{P|Q}$ can have modal depth much greater than $\max(\chi_P, \chi_Q) + 1$, then bounds routed through characteristic formulae degrade logarithmically. The tower-routed bounds are unaffected, but the question should be settled before anyone builds on the other route.
3. **The copy series under merger.** [4, §8] shows that merger sets $\beta_1 = 0$ and destroys factorisation. What does it do to $\text{CN}^\bullet$? Our expectation is that it collapses two adjacent strata into one, which would make the series a rather direct diagnostic for whether a composite has merged or merely composed — and would give [3]’s composition/merger distinction an observable signature.
4. **Affordable scope and the individuation fixed point.** Definition 3.12 makes affordable scope depend on a budget; [4, Rem. 5.24] makes individuation a fixed point. Both are circular in the same way — the region determines what can be afforded and what can be afforded determines the region. Do they have a common solution, and is the individual the least one?
5. **The survey is an assay.** A survey visits channels and decides bisimilarity at depth n ; that is an experiment, and by [1, §4] an experiment has three outcomes, the third being budget-relative. So a survey has a $\perp$ column, and a copy number is properly a triple: confirmed copies, confirmed non-copies, and channels the budget could not settle. What is the right generalisation of Definition 4.1 that carries the third column, and does the biosignature criterion survive it?
6. **Estimating the inflation.** Remark 4.7 gives a mechanism for systematic overcounting in biosignature detection without giving its size. What is the effective resolution of a mass spectrometer, expressed as a modal depth over the reaction system it observes, and how deep do the kinds in a prebiotic soup separate?
7. **Generator length as a measurable.** h/ℓ is proposed in Remark 6.3 as “how much depth a description buys”. For organisms, ℓ has an obvious candidate proxy and h does not. Is there a way to estimate the number of levels of *owned* inside an organism has, independently of anatomy?
8. **The general additivity question.** Proposition 5.4 needs disjoint rooting. Ecologies that arose rather than were designed will have overlapping strata. Is there a bound on the collapse — some inclusion–exclusion over the overlap grades of [3, §6] — or does the sum fail catastrophically?

A Rholang

Two contracts. The first decides membership in a generated scope by descent, per Proposition 3.9. The second surveys a scope and returns a copy series. Both follow the conventions of [2, App. A]:

a bare parameter is a name, an @-pattern binds data, and a channel is passed as *c. Guards marked `///!` ideal stand for decision procedures we assume rather than implement.

```

new InScope, Split, Survey, Series, BisimN, Atom in {

// ---- membership by descent -----
// mu X . @( (phi \ / X) | (psi \ / X) ), presented as a list of
// atomic predicates and tested by structural descent on the quote.

contract InScope(@nm, atoms, ret) = {
  // atoms is a list of atomic predicate channels; nm is the name
  // under test. Stratum 0: the drop satisfies some atom.
  new hit in {
    Atom!(nm, *atoms, *hit) |
    for (@isAtom <- hit) {
      if (isAtom) { ret!(true) }
      else {
        // Stratum j+1: the drop is a parallel composition, and by
        // Prop. "unique decomposition" the split is unique, so no
        // search is required -- one split, two recursive calls.
        new sp in {
          Split!(nm, *sp) |
          for (@p, @q, @ok <- sp) {
            if (not ok) { ret!(false) }
            else {
              new l, r in {
                InScope!(p, *atoms, *l) |
                InScope!(q, *atoms, *r) |
                for (@a <- l & @b <- r) { ret!(a and b) }
              }
            }
          }
        }
      }
    }
  }
}

// Split returns the unique decomposition of a quoted parallel
// composition. Termination of the recursion above is exactly the
// no-self-code theorem: p and q code strictly smaller terms.
contract Split(@nm, ret) = {
  match *nm {
    { P | Q } => { ret!(@P, @Q, true) } ///! ideal
    _ => { ret!(Nil, Nil, false) }
  }
}

// ---- the survey -----
// Visit each channel of the scope, decide n-step bisimilarity with
// the target, and bin the result by stratum. One metered
// rendezvous per visit: this contract is where a copy number is
// paid for.

contract Survey(@target, chans, @n, ret) = {
  new tally in {
    tally!({}) |
    for (@c <= chans) { // one visit per channel
      new b, s in {
        BisimN!(c, target, n, *b) | ///! ideal, cost kappa(n)
        Stratum!(c, *s) |
        for (@same <- b & @j <- s) {
          for (@acc <- tally) {
            if (same) { tally!(acc.set(j, acc.getOrElse(j,0) + 1)) }
            else { tally!(acc) }
          }
        }
      }
    }
  }
}

```

```

    }
  } |
  for (@acc <- tally) { ret!(acc) } // the copy series
}
} |

// The flat collapse, for comparison with the received definition.
contract Series(@series, ret) = {
  ret!( series.values().fold(0, (a, b) => a + b) )
}
}

```

Three things the listing makes visible that the prose does not. The recursion in **InScope** has no fixed depth bound in its own code — it terminates because of a theorem about the calculus, not because of a counter. **Survey** charges once per channel, which is what makes affordable scope a budget question rather than a definitional one. And **Series** exists only to throw information away; it is the received copy number, written as the lossy projection it is.

B Computations

All figures in this note come from `scope.py`, in four blocks.

(1) Extension growth. Names of stratum $\leq j$ in the two-atom generated scope, counted up to structural congruence, satisfy $u_0 = 2$ and $u_{j+1} = 2 + \binom{u_j+1}{2}$, since the composites at stratum $\leq j+1$ are unordered pairs with repetition drawn from u_j and parallel composition is commutative. Values in Table 1: 2, 5, 17, 155, 12 092, 73 114 280. The generator is three symbols wide throughout.

(2) Overcounting. $CN_n = b^{h-n}$ for the behaviour-tree population of Proposition 4.4. At $b = 3$, $h = 8$: 6561, 2187, 729, 243, 81, 27, 9, 3, 1 for $n = 0, \dots, 8$. Plotted as Figure 2.

(3) Sampling floor. $\lceil mp^{-d} \rceil$ at $m = 20$, tabulated in Table 3 over six reliabilities from 0.99 down to 0.50 and depths d from 1 to 8. The optimal depth column of Table 4 maximises $Y(d)p^d - cd$ with $Y(d) = 100(1 - 2^{-d})$, $c = 2$, reproducing the d^* values of [4, App. B]: 5 at $p \geq 0.99$, 3 at $p \in \{0.95, 0.90\}$, 2 at $p = 0.75$.

(4) The composite. Four skills with the parameters of Table 4; 96 learners per individual; twelve individuals; copy series (1152, 48, 12, 1) against a flat collapse of 1213; assembly floor 28 shared against 43 unshared, a saving of 15 (34.9%), of which 16 of the 28 (57.1%) is content. Individuation at $\theta = 0.2$, $\rho = 5$: the individual has $\varepsilon\theta/\rho = 4 \times 10^{-4}$ and $r^* = 1250$; the pod has 4×10^{-3} and $r^* = 75$. Observer reports by resolution in Table 6.

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